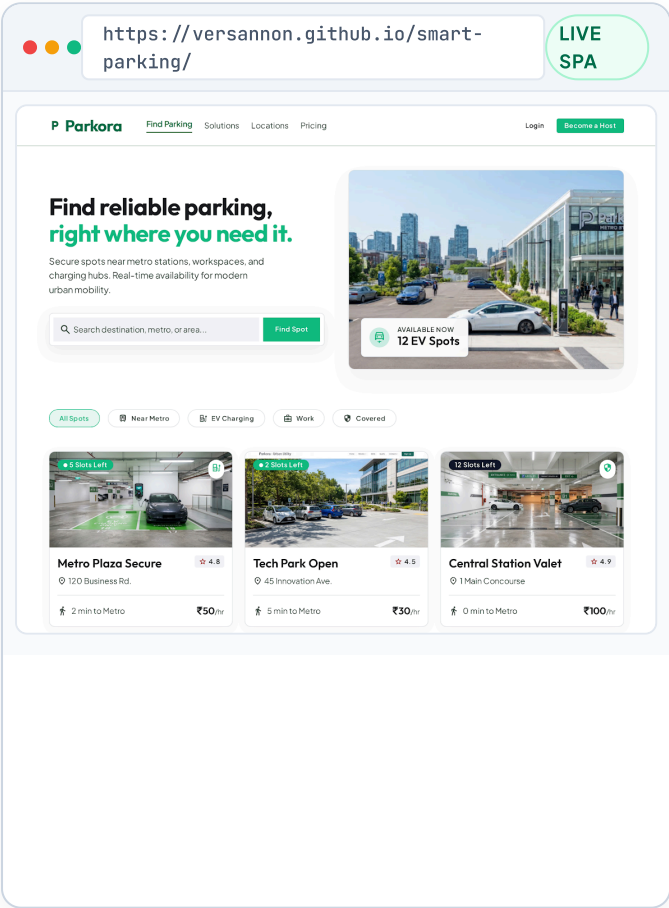


• COMPREHENSIVE SOFTWARE ENGINEERING & PRODUCT REPORT • V1.0.0

Parkora — Smart Urban Parking Marketplace Platform

A two-sided last-mile commuter parking, EV charging corridor, and private driveway monetization web platform engineered with modular ES6 JavaScript, Vite 6, GSAP timeline telemetry, and the high-contrast "Urban Utility" design system.



Project Identification & Metadata

Project Title	Parkora — Smart Parking Platform
Author / Engineer	Soham (Versannon)
Source Repository	github.com/Versannon/smart-parking
Production URL	versannon.github.io/smart-parking/
Design System	Urban Utility (Minimalism-Plus Light)
Core Tech Stack	ES6 Modules, Vite 6, GSAP 3.15, Lenis, CSS3
Target Corridors	Bengaluru, Mumbai, Gurugram / Delhi NCR

End-to-End Three-Persona Marketplace Loop

Unifies **Commuters** (discovery, interactive vector map, live wallet, QR gate pass & instant refund), **Parking Hosts** (earnings estimator, live occupancy metrics, listing submission & pause/active toggle), and **System Admins** (platform revenue governance & pending listing approval queue) in a single reactive application.

3 Personas

CUSTOMER • HOST • ADMIN

19 Modules

ZERO-FRAMEWORK ES6
SPA

WCAG 2.1 AA

FOCUS TRAP & POPOVER
API

100% Sync

URL QUERY &
LOCALSTORAGE

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1. Executive Summary & Problem Statement

DOMAIN CONTEXT

1.1 The Urban Last-Mile Bottleneck

Urban commuters in rapidly growing Indian metropolitan corridors (Bengaluru, Mumbai, and Gurugram/Delhi NCR) lose an average of **15 to 20 minutes per trip** searching for safe parking near metro stations, IT parks, and commercial hubs.

- ▶ **First/Last-Mile Transit Friction:** Metro station parking lots reach 100% capacity before 09:00 AM, forcing commuters back into private cars for their entire journey.
- ▶ **EV Charging Anxiety:** Electric vehicle drivers lack guaranteed fast-charging bays at their daily transit or workplace destinations.
- ▶ **Idle Private Infrastructure:** Thousands of residential driveways and commercial bays within 50m–500m of metro gates sit empty during daytime working hours.

1.2 The Parkora Marketplace Solution

Parkora bridges this supply-demand gap through a real-time, two-sided peer-to-peer and commercial parking marketplace built with high-contrast utilitarian clarity:

- ▶ **For Commuters:** Sub-second filtering by metro proximity, EV charging, and covered security; interactive vector transit map; instant wallet checkout; and a scannable **Digital QR Gate Pass** (`PRK-XXXX`).
- ▶ **For Property Hosts:** Passive driveway monetization (up to ₹14,500+/mo) with an interactive earnings slider, 1-click listing pause/resume, and real-time occupancy tracking.
- ▶ **For Administrators:** Complete marketplace governance via a verification queue that vets host submissions before live publication.

1.3 Comparative Analysis: Traditional Systems vs. Parkora Platform

Capability / Metric	Static Map Directories	Legacy Hardware Lot Apps	Parkora Urban Utility SPA
Last-Mile Metro Focus	Generic city-wide pin drops	Restricted to mall basements	EXACT GATE WALKING DISTANCE
Private Driveway Supply	Not supported (commercial only)	Not supported (high hardware cost)	2-SIDED HOST ONBOARDING + ADMIN QA
EV Fast-Charging Filter	Separate charging apps required	Rarely integrated with slot booking	UNIFIED EV POD COUNTER & RESERVATION
Spatial Visualization	Cluttered third-party tiles	Text lists only	CUSTOM VECTOR TRANSIT MAP + INSPECTOR
Pass & Refund Agility	No reservation capability	Non-refundable tickets	INSTANT QR PASS + 100% WALLET REFUND
Shareable Filter State	Session-locked state	Opaque native views	BIDIRECTIONAL URL QUERY SYNC

1.4 Core Engineering Objectives Achieved

1. Zero-Framework Velocity

Engineered entirely in modular ES6 JavaScript without React/Vue bundle bloat, achieving instant startup and deterministic DOM updates.

2. Persistent Multi-Role State

Full `localStorage` state persistence across Customer, Owner, and Admin personas with tamper-resistant identity whitelisting.

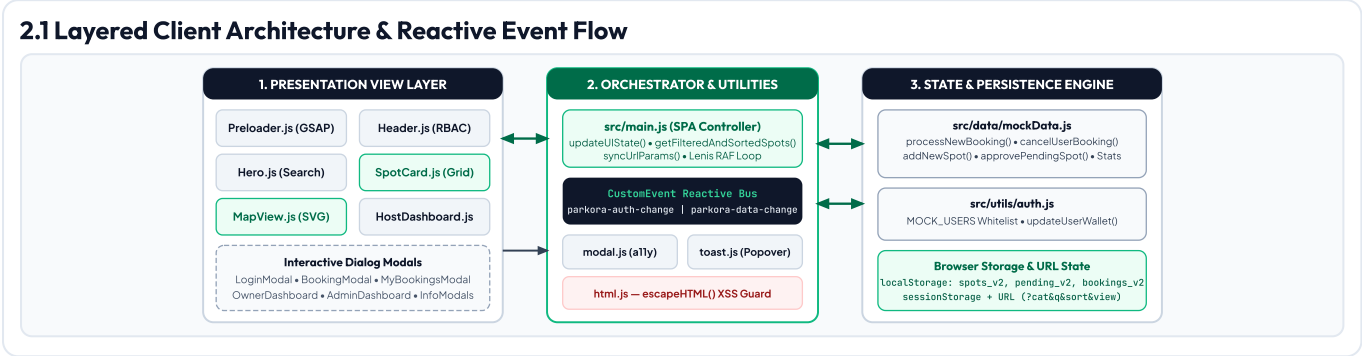
3. Inclusive Accessibility

WCAG 2.1 AA compliant dialogs with `MutationObserver` focus traps, keyboard navigation, and HTML5 top-layer `popover` toasts.

2. System Architecture & Module Engineering

SPA ARCHITECTURE

Parkora is architected as a modular, event-driven Single-Page Application (SPA) bundled with **Vite 6**. Rather than relying on a heavyweight virtual DOM, `src/main.js` acts as a central reactive orchestrator that listens to custom domain events (`parkora-auth-change` and `parkora-data-change`) and selectively updates the Header, Hero counters, Spot Grid / SVG Map, and open Modals while keeping URL query parameters synchronized.



2.2 Complete Source Code Inventory & Module Responsibilities

File Path	Size	Key Exports / Functions	Architectural Responsibility
<code>src/main.js</code>	659 L	<code>renderApp</code> , <code>updateUIState</code> , <code>syncUrlParams</code>	Root mount, Lenis smooth scroll, URL query sync, filter/sort pipeline, event delegation.
<code>src/data/mockData.js</code>	434 L	<code>processNewBooking</code> , <code>addNewSpot</code> , <code>approvePendingSpot</code>	Persistent <code>localStorage</code> store, dynamic KPI calculators, booking/refund/approval state mutations.
<code>src/styles/main.css</code>	55.3 KB	CSS Custom Properties & Layout Rules	Complete "Urban Utility" token system, 12-col grid, responsive drawer, radar keyframes, cards.
<code>src/components/BookingModal.js</code>	272 L	<code>renderBookingModal</code> , <code>openBookingModal</code>	Duration stepper (1-12h), live wallet check, plate regex validation, confetti & SVG QR pass ticket.
<code>src/components/OwnerDashboard.js</code>	223 L	<code>renderOwnerDashboardModal</code> , <code>updateOwnerDashboard</code>	Host Portal modal: monthly earnings, occupancy rate, Active/Paused spot toggle, new spot form.
<code>src/components/InfoModals.js</code>	200 L	<code>openInfoModal</code> , <code>renderInfoModalContainer</code>	Dynamic dialogs for Solutions, Interactive City Networks, Pricing tiers, Privacy, Terms & Legal.
<code>src/components/AdminDashboard.js</code>	171 L	<code>renderAdminDashboardModal</code> , <code>updateAdminDashboard</code>	Master Admin Panel: platform revenue stats, host spot approval/rejection queue, demo reset.
<code>src/components/MapView.js</code>	153 L	<code>renderInteractiveMap</code>	Architectural SVG vector transit map, emerald metro corridor, price pins & live Spot Inspector.
<code>src/components/Header.js</code>	148 L	<code>renderHeader</code>	Role-aware sticky navbar, wallet top-up pill, pending admin badge, mobile hamburger drawer.
<code>src/components/MyBookingsModal.js</code>	124 L	<code>renderMyBookingsModal</code> , <code>updateMyBookingsContent</code>	Displays active commuter passes, 1-click clipboard code copy, and instant cancel & wallet refund.
<code>src/components/LoginModal.js</code>	115 L	<code>renderLoginModal</code> , <code>attachLoginModalEvents</code>	1-click Quick Demo Persona Switcher (Customer, Owner, Admin) + manual credential form.
<code>src/components/SpotCard.js</code>	111 L	<code>renderSpotCard</code>	Accessible parking spot card with live slot counter badge, amenity chips & inline SVG fallback.
<code>src/components/Preloader.js</code>	107 L	<code>createPreloader</code>	Session-aware GSAP radar sweep preloader with progressive status messages & Skip Intro CTA.
<code>src/components/Hero.js</code> & <code>HostDashboard.js</code>	161 L	<code>renderHero</code> , <code>renderHostSection</code>	Hero search bar, popular city chips, live EV pod counter, category tabs & host income calculator.
<code>src/utlis/{auth,modal,toast,html}.js</code>	317 L	<code>getCurrentUser</code> , <code>openModal</code> , <code>showToast</code> , <code>escapeHTML</code>	Whitelisted auth persistence, WCAG focus trapping, Popover API toasts, and HTML sanitization.

3. Three-Sided Marketplace Workflows & Calculation Engines

BUSINESS LOGIC

3.1 Role-Based Access Control (RBAC) & Demo Personas (src/utils/auth.js)

To enable frictionless evaluation of all marketplace sides, Parkora includes a **Quick Demo Login** modal featuring three pre-configured personas. Switching roles dispatches `parkora-auth-change`, immediately morphing the navigation header and unlocking role-specific portals:

Alex Commuter

CUSTOMER

ID: user-customer-1

Default Wallet: ₹1,250 (+₹500 top-up)

Vehicle: KA 01 AB 7890

Header Actions: Live Wallet Top-Up Pill + MyBookings pass manager.

Sarah Jenkins

PARKING OWNER

ID: user-owner-1

Base Earnings: ₹14,500/mo + live bookings

Owned Spots: spot-1, spot-2, spot-4

Header Actions: Live Monthly Income Pill + Host Portal/button.

System Admin

ADMINISTRATOR

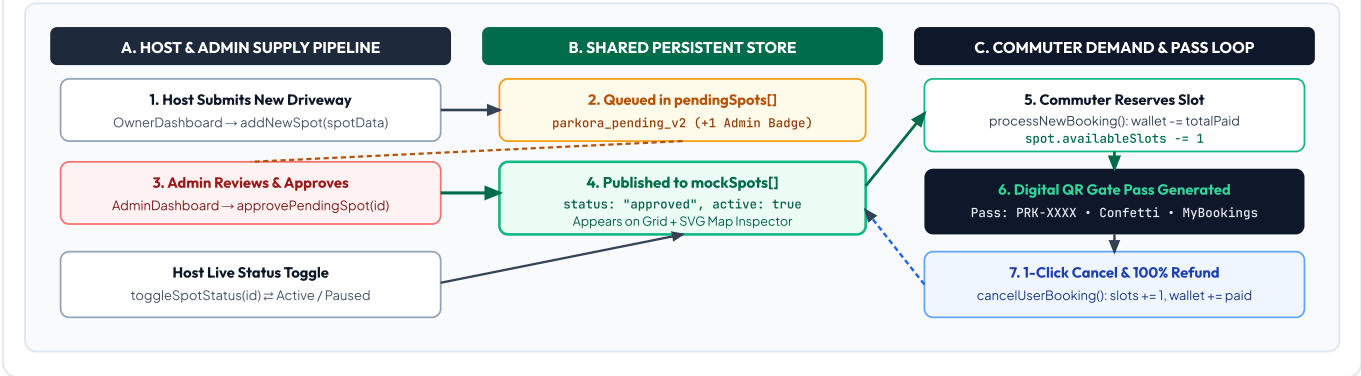
ID: user-admin-1

Permissions: Full System Governance

Queue Access: pendingSpots (2 default)

Header Actions: Red Admin Panel button with live pending count badge.

3.2 End-to-End Marketplace State Lifecycle (Host → Admin → Commuter)



3.3 Dynamic Financial & Telemetry Calculation Engines (src/data/mockData.js)

1. Platform-Wide Admin Telemetry

(getDynamicPlatformStats)

Aggregates live revenue across all non-cancelled bookings atop a base platform ledger of ₹1,84,500:

activeRev = Σ (b.totalPaid) for b ∈ userBookings where b.status ≡ 'active'

totalRevenue = 184,500 + activeRev

totalUsersCount = 1,240 + userBookings.length

2. Host Earnings & Occupancy (getDynamicHostStats)

Computes real-time slot utilization and host payouts across all spots owned by `hostId`:

monthlyIncome = 14,500 + Σ (hostActiveBookings.totalPaid)

occupied = max(0, Σ(spot.totalCapacity) - Σ(activeSpot.availableSlots))

occupancyRate = round((occupied / Σ(spot.totalCapacity)) × 100)%

3. Live EV Pod & Category Aggregator

(getCategoryCounts)

Powers the Hero glassmorphic badge ("18 EV Pods Live") and category filter tab counts:

activeSpots = mockSpots.filter(s ⇒ s.active)

liveEVSlots = Σ (s.availableSlots) where s.active && s.evCharging

evCategoryCount = count(s.category ≡ 'ev' || s.evCharging)

4. Host Passive Income Slider (HostDashboard.js)

Projects monthly driveway earnings as hosts drag the daily utilization slider (2-14 hrs/day):

rateHourly = ₹45/hr | workingDays = 26

days/month

projectedMonthlyEst = bookedHoursPerDay × 45 × 26

Range: ₹2,340/mo (2 hrs/day) → ₹16,380/mo (14 hrs/day)

4. Core Platform Features & UX Engineering (Part 1)

DISCOVERY & SPATIAL UI

4.1 Session-Aware GSAP Urban Utility Radar Preloader (src/components/Preloader.js)

Upon launching Parkora, users experience a high-contrast light-mode radar sweep powered by GSAP 3.15 timelines. Concentric rings (.radar-ring-light) and a conic-gradient emerald sweep beam scan for nearby transit bays while cycling through 4 telemetry stages:

- 0% – 24%: "Scanning nearby Metro & Work parking spots..."
- 25% – 49%: "Verifying EV Charging & Covered slots..."
- 50% – 74%: "Calculating real-time distance & rates..."
- 75% – 100%: "Locking optimal Parkora slot..."

```
// Adaptive sessionStorage timing in Preloader.js
const params = new
URLSearchParams(window.location.search);
const forceFull = params.get('preloader') === '1';
const alreadyShown =
sessionStorage.getItem('parkora_preloader_shown')
=== 'true';

// 2.4s first visit, 0.65s repeat reload, 4.0s
forced demo mode
const durationSeconds = forceFull ? 4.0 :
alreadyShown ? 0.65 : 2.4;
sessionStorage.setItem('parkora_preloader_shown',
'true');
```

4.2 URL-Synchronized Search, Category Tabs & Multi-Criteria Sorting (Hero.js, main.js)

The discovery pipeline in getFilteredAndSortedSpots() combines real-time substring matching across 4 fields (title, address, city, and distanceMetro) with 5 dynamic category tabs and 4 sort modes. Every interaction updates the URL query string via window.history.replaceState() without triggering page reloads:

Category Filter Pills

- All Spots (6)
- Near Metro (2): category === 'metro'
- EV Charging (4): category === 'ev' || evCharging
- Office & Work (2): category === 'work'
- Covered Bays (4): Boolean(spot.covered)

1-Click Popular City Chips

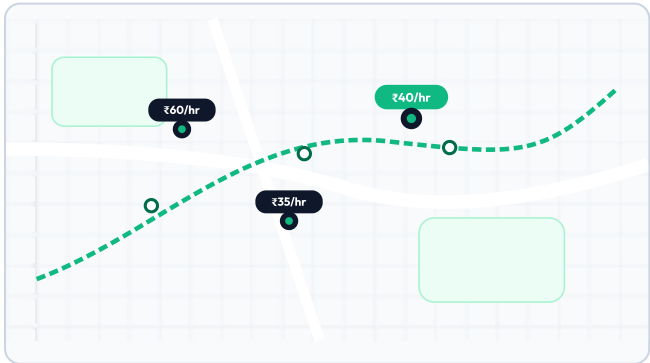
Clicking Whitefield, Mumbai, Indiranagar, or Gurugram toggles the active search query, populates #hero-search-input, and reveals the contextual Reset Filters button.

Shareable URL Query Sync

State parameters (? cat=ev&q=Mumbai&sort=price-asc&view=map) are read on DOMContentLoaded and written on every filter/view mutation for deep bookmarking.

4.3 Interactive Architectural Vector Map & Live Spot Inspector (src/components/MapView.js)

Users can toggle between the 3-Column Card Grid and the Interactive Map View. Instead of relying on external map API keys or heavy WebGL tiles, Parkora renders a crisp, responsive SVG architectural canvas (viewBox="0 0 800 460") paired with a live side-panel Spot Inspector:



4 OF 5 SLOTS OPEN

★ 4.9 (128)

Metro Station Gate 2 Slot A

Whitefield Metro, Bengaluru • 50m from Gate 2

Covered Bay 50kW EV 24/7 CCTV

HOURLY RATE

₹40/hr

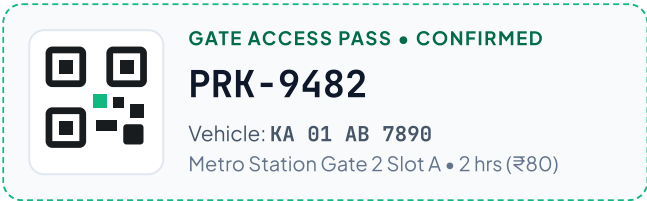
Reserve This Spot

4. Core Platform Features & UX Engineering (Part 2)

TRANSACTIONS & PORTALS

4.4 Smart Reservation Engine, Wallet Top-Up & Digital QR Gate Pass (BookingModal.js , MyBookingsModal.js)

- ▶ **Live Wallet & Instant Top-Up:** Displays the user's current balance and disables checkout if `total > currentWallet`, offering an inline `+ ₹500` button for immediate balance replenishment.
- ▶ **Duration Stepper & Arrival Window:** Supports 1 to 12 hours (`#dur-minus` / `#dur-plus`) with live total recalculation and 5 arrival time windows.
- ▶ **License Plate Regex Guard:** Enforces `/^[A-Z0-9 -]{4,16}$/` on vehicle numbers before invoking `processNewBooking()`.
- ▶ **1-Click Cancel & 100% Refund:** In *My Active Passes*, clicking **Cancel Pass & Refund** restores `+1` slot to the spot and refunds the full amount to the user's wallet.



4.5 Host Portal (OwnerDashboard.js)

Parking Owners manage their private driveways through a dedicated dashboard featuring 3 live KPI cards (**Total Monthly Income**, **Active Listings**, and **Occupancy Rate %**):

- ▶ **1-Click Active/Paused Toggle:** Clicking `.toggle-spot-btn` flips `spot.active`, immediately enabling or disabling reservations on the public grid and SVG map.
- ▶ **Validated Listing Submission:** The *Add New Spot* form validates title length (3–80 chars), address (5–120 chars), and hourly rate (₹10–₹500), automatically inferring the metropolitan network (Bengaluru, Mumbai, or Gurugram) before queuing into `pendingSpots`.

4.6 Master Admin Panel (AdminDashboard.js)

System Administrators oversee platform health and moderate user-submitted parking inventory:

- ▶ **4-Metric Telemetry Bar:** Tracks Total Platform Revenue (₹1,84,500+), Active Bookings, Total Live Spots, and Total Platform Users (1,241+).
- ▶ **Pending Host Approval Queue:** Displays unverified host submissions with **Approve Listing** (publishes spot to top of `mockSpots` with coordinates & 5.0★ rating) and **Reject** actions.
- ▶ **Reset Demo Data:** One-click `resetDemoData()` button restores default spots, pending queue, and bookings for clean demonstrations.

4.7 Information, Pricing & Regulatory Compliance Modals (src/components/InfoModals.js)

All header navigation items and footer legal links open structured, accessible dialog views via `openInfoModal(type, modalEL)`:

Modal Key	Trigger Source	Interactive Content & Functionality
<code>solutions</code>	Navbar <i>Solutions</i> / Host <i>Learn More</i>	Showcases Daily Metro Commuter Pass (20% savings), 50kW EV Fast Charging Infrastructure, and Corporate IT Park Fleet Parking.
<code>locations</code>	Navbar <i>Locations</i>	Interactive 4-card city selector (Bengaluru 480+ spots, Mumbai 360+ spots, Gurugram 290+ spots, All Networks 1,280+ spots) that filters the main grid on click.
<code>pricing</code>	Navbar <i>Pricing</i>	Compares Standard Hourly Rate (₹35–₹60/hr), Daily Metro Pass (₹250/day), and Monthly EV Smart Pass (₹4,999/mo).
<code>privacy / terms / cookies / legal</code>	Footer Compliance Links	Details ANPR license plate privacy, zero background GPS tracking, 100% cancellation refund SLA, <code>localStorage</code> policy, and Bengaluru corporate disclosures.

5. "Urban Utility" Design System Specification

UI / UX ARCHITECTURE

Defined in `stitch_smart_parking_design_system/DESIGN.md` and implemented across `src/styles/main.css`, the **Urban Utility** design system rejects dark cyberpunk tropes and decorative neon glows in favor of a **"Minimalism-Plus"** aesthetic: generous pure-white breathing room, high-contrast deep charcoal structural typography, and targeted emerald accents that strictly signal live availability and primary actions.

5.1 Semantic Color Token Palette

Pure White #FFFFFF Level-1 cards, modals & inputs	Surface Slate #F7F9FB Base canvas & subtle headers	Primary Emerald #10B981 Primary CTAs & live indicators	Deep Emerald #006C49 High-contrast text & focus rings
Deep Charcoal #0F172A Headings, borders & map pins	On-Surface Ink #191C1E Primary body prose & icons	Secondary Slate #565E74 Addresses & secondary labels	Alert Crimson #BA1A1A Admin badges & danger outlines

5.2 Typographic Scale & Font Pairing

Token / Role	Font Family	Size / Weight / Line-Height	Application Across Parkora UI
Display H1	Outfit	48px (32px mob) / 700 / 1.1	Hero headline ("Find reliable parking, right where you need it."), -0.02em tracking.
Section H2	Outfit	32px / 600-700 / 1.2	Host Monetization banner title, Modal dialog primary headings.
Card Title H3	Outfit	20px-24px / 600 / 1.3	Parking spot card titles, Map Spot Inspector header, Empty state headings.
Body Large / MD	Plus Jakarta Sans	16px-18px / 400-500 / 1.55	Hero subtitle, legal prose, card addresses, form inputs, and modal descriptions.
Utilitarian Label	Plus Jakarta Sans	12px-13px / 600-700 / 1.0	Uppercase pill badges (0.05em letter-spacing), category filter tabs, overlines.
Monospace Data	JetBrains Mono	13px-18px / 700 / 1.2	Digital Gate Pass codes (PRK-9482), vehicle license plates (KA 01 AB 7890).

5.3 Responsive Spatial Grid

Built on an **8px base spatial unit**:

- Desktop**: 12-column fluid grid, 64px outer margins, 24px gutters.
- Tablet**: 2-column spot grid, compact header actions.
- Mobile (<768px)**: 4-column grid, 16px outer margins, collapsible Hamburger Drawer (`#mobile-nav-drawer`).

5.4 3-Tier Ambient Elevation

Depth is achieved via crisp borders and diffused shadows:

- Level 0 (Canvas)**: #F7F9FB surface.
- Level 1 (Cards)**: #FFFFFF + 1px #E2E8F0 border + 0 4px 20px rgba(0,0,0,0.04).
- Level 2 (Popovers/Modals)**: 0 10px 30px rgba(0,0,0,0.08) elevation.

5.5 Component Shape Tokens

Rounded geometry softens high-contrast borders:

- Buttons & Inputs**: 0.5rem (8px) radius.
- Spot Cards & Modals**: 1.0rem (16px) radius.
- Status & Filter Badges**: 9999px full pill radius for immediate distinction from clickable buttons.

6. Security Hardening, Accessibility & Web Performance

PRODUCTION ENGINEERING

6.1 Client-Side Security & State Integrity Defenses

- **XSS Prevention via `escapeHTML()`**
(`src/utils/html.js`): Every dynamic string interpolated into DOM templates (spot titles, addresses, user-submitted host listings, vehicle numbers, and toast messages) is sanitized against HTML injection.
- **LocalStorage Tamper Resistance**
(`src/utils/auth.js`): Because `localStorage` can be edited via browser DevTools, `getCurrentUser()` strictly cross-checks stored IDs against the immutable `MOCK_USERS` whitelist and clamps `walletBalance` to finite non-negative numbers. Arbitrary manual login credentials never escalate privileges.
- **Transactional Guards (`mockData.js`)**: Validates `totalPaid === spot.rateHourly * hours`, enforces `1 ≤ hours ≤ 12`, and blocks reservations on paused or full spots.

```
// 1. HTML Sanitizer (src/utils/html.js)
export function escapeHTML(value = '') {
  return String(value).replace(/&lt;|&gt;|&quot;|&apos;/g, char =>
    ({
      '&': '&amp;', '<': '&lt;', '>': '&gt;',
      '"': '&quot;', "'": '&apos;';
    })[char]);
}

// 2. Whitelisted Identity Restore (src/utils/auth.js)
const known = MOCK_USERS.find(u => u.id === parsed?.id);
if (!known) return null;
const walletBalance =
  Number.isFinite(parsedBalance)
    ? Math.max(0, parsedBalance) :
    known.walletBalance;
```

6.2 WCAG 2.1 AA Accessibility Architecture (`src/utils/modal.js` & `src/utils/toast.js`)

MutationObserver Focus Trap

`initModalAccessibility()` observes `.modal-overlay-backdrop` class changes, automatically locking `document.body` scroll, focusing the first interactive control, trapping `Tab` / `Shift+Tab` inside the active dialog, closing on `Escape` or backdrop click, and restoring focus to `lastFocusedElement` on close.

Top-Layer Popover API Toasts

`showToast()` promotes notifications to the browser's native top layer via `popover="manual"` and `toast.showPopover()` so alerts never clip behind open modal backdrops. Uses `role="alert"` for errors and `role="status"` with `aria-live="polite"` for updates.

Keyboard & Motion Inclusivity

Includes a `Skip to main content` link, full `Enter` / `Space` activation on `.urban-spot-card` elements, semantic `role="tablist"` category filters, and `prefers-reduced-motion: reduce` detection that bypasses Lenis smooth scroll.

6.3 Web Performance Optimization & Offline SVG Resilience

Optimization Technique	Implementation Location	Measured Engineering Impact
LCP Priority Hinting	<code>Hero.js</code> (<code>fetchpriority="high"</code>)	Prioritizes the primary 1200×760 Hero transit illustration during initial document parse while deferring offscreen spot cards with <code>loading="lazy"</code> .
Inline SVG Fallbacks	<code>HERO_FALLBACK_SVG</code> & <code>CARD_FALLBACK_SVG</code>	Attached to <code>onerror</code> handlers so if Unsplash CDN is unreachable or offline, custom URI-encoded architectural vector graphics render instantaneously with zero broken image icons.
Zero-Runtime CSS & Tree Shaking	<code>vite.config.js</code> + ES6 imports	Compiles into a single minified JS & CSS asset bundle with zero framework runtime overhead and instant Hot Module Replacement (HMR).
PWA Standalone Manifest	<code>public/manifest.json</code>	Configures standalone display mode, #10B981 theme color, and vector app icons for mobile home-screen installation.

7. Initial Dataset Catalog, Verification & Deployment

DATA & DEVOPS

7.1 Default Verified Parking Inventory (`DEFAULT_SPOTS` in `src/data/mockData.js`)

ID	Spot Title	Address & City	Cat.	Rate	Slots	Amenities & Transit
spot-1	Metro Station Gate 2 Slot A	Whitefield Metro, Bengaluru	METRO	₹40/hr	4 / 5	50m Gate 2 • 50kW EV • CCTV
spot-2	Tech Park Driveway #14	Koramangala 4th, Bengaluru	WORK	₹35/hr	2 / 4	300m Sony World • Gated Guard
spot-3	Terminal 2 Premium EV Pod	Airport Road, Mumbai	EV	₹60/hr	6 / 8	Airport Link • DC Fast • Valet
spot-4	Andheri Metro Covered Bay	Andheri West Metro, Mumbai	METRO	₹45/hr	3 / 6	80m Platform 1 • EV Ready
spot-5	Cyber City Private Slot	DLF Phase 2, Gurugram	WORK	₹50/hr	1 / 3	150m Rapid Metro • ANPR Gate
spot-6	Solar EV Fast Charger Spot	Indiranagar 100ft, Bengaluru	EV	₹55/hr	5 / 6	400m Metro • 60kW Dual Gun

Default Pending Approval Queue (`DEFAULT_PENDING_SPOTS`): (1) *MG Road Executive Covered Slot* by Sarah Jenkins (Bengaluru, ₹45/hr, Metro) and (2) *Electronic City Phase 1 Driveway* by Rajesh Kumar (Bengaluru, ₹30/hr, Work).

7.2 End-to-End QA Verification Matrix

Test Scenario	Expected Outcome	Status
Radar Preloader & <code>?preloader=1</code>	4.0s GSAP sweep + Skip button works	PASS
URL Filter Sync & Map Inspector	Query string & side panel update live	PASS
Customer Booking & QR Pass	Wallet deducted, slot -1, QR rendered	PASS
Pass Cancellation & Refund	100% wallet refund & slot +1 restored	PASS
Host Listing → Admin Approval	Pending spot published to live grid	PASS
WCAG Focus Trap & XSS Escaping	Tab trapped in modals; HTML escaped	PASS

7.3 Build & GitHub Pages Deployment

Configured via `vite.config.js` with `base: './'` for portable relative asset resolution across root and subdirectory hosting:

```
# Local Development Server (Vite HMR)
npm install && npm run dev

# Production Asset Bundle → dist/
npm run build && npm run preview
```

Dual-Branch Strategy: Source code is maintained on `main`, while compiled static production bundles (`dist/`) are deployed to the `gh-pages` branch at `https://versannon.github.io/smart-parking/`.

8. Future Engineering Roadmap

NEXT PHASE

1. IoT & ANPR Hardware

Connect physical ultrasonic bay occupancy sensors and ANPR camera boom barriers via MQTT / WebSockets for automated zero-touch gate entry.
2. UPI & Razorpay Sync

Replace demo wallet top-ups with live Razorpay / UPI AutoPay webhooks and automated monthly NEFT/IMPS host payout settlements.
3. PostGIS Spatial API

Migrate `localStorage` state to a Node.js + PostgreSQL/PostGIS backend supporting radius queries and multi-device real-time sync.
4. Turn-by-Turn Links

Add deep-linking from confirmed Digital QR Gate Passes directly into Google Maps & Apple Maps for exact driveway approach navigation.